RAJALAKSHMI ENGINEERING COLLEGE

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| **Started on** | Thursday, 13 August 2026, 3:53 PM |
| **State** | Finished |
| **Completed on** | Thursday, 13 August 2026, 3:57 PM |
| **Time taken** | 4 mins 36 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

**Note:** No need of counter increment for declarations and scanf() and counter variable printf() statement.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:**

```
#include <stdio.h>
void factor(int num) {
    int count = 0;
    count++; // initialization of i = 1
    for(int i = 1 ; i <= num ; i++) {
        count++; // for the times of valid condition checks
        count++; // for i++
        if(num % i == 0) {
        count ++; // for the count of times of execution of if condition
        //printf("%d ", i);
            }
}
        printf("%d", count);
        }
        int main() {
            int n;
            scanf("%d", &n);
            factor(n);
            return 0;

        }
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 12 | 31 | 31 | ✔ |
| ✔ | 25 | 54 | 54 | ✔ |
| ✔ | 4 | 12 | 12 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course